

# Relations

## Chapter 9

### Chapter Summary

- Relations and Their Properties
- Representing Relations
- Closures of Relations
- Equivalence Relations
- Partial Orderings

# Relations and Their Properties

Section 9.1

## Section Summary

- Relations and Functions
- Properties of Relations
  - Reflexive Relations
  - Symmetric and Antisymmetric Relations
  - Transitive Relations
- Combining Relations

## Social Relationships

- There are many kinds of relationships in the world:
- Relative: Relationship by blood or by a common ancestor.
- Friendship: boyfriend and girlfriend
- Relations between Teachers and students
- Relations between bosses and employees

## Social Relationships

- Relations between war and peace
- Relations between city and village
- Relations between God and mankind
- Relations between mankind and their environment
- Relations between obama and osama (bin laden)
- And so on...

## Abstract Relationships

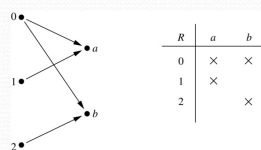
- The question is how to represent relationship in mathematical methods
- N-ary relationships (complex ): relationships among many objects.
- But most of the relationship can be formalized in the idea of binary relation.
- Binary relation is the simplest relation, it is what we will study in this course.

## Binary Relations

**Definition:** A *binary relation*  $R$  from a set  $A$  to a set  $B$  is a subset  $R \subseteq A \times B$ .

**Example:**

- Let  $A = \{0,1,2\}$  and  $B = \{a,b\}$
- $\{(0, a), (0, b), (1, a), (2, b)\}$  is a relation from  $A$  to  $B$ .
- We can represent relations from a set  $A$  to a set  $B$  graphically or using a table:



Relations are more general than functions. A function is a relation where exactly one element of  $B$  is related to each element of  $A$ .

## Binary Relation on a Set

**Definition:** A binary relation  $R$  on a set  $A$  is a subset of  $A \times A$  or a relation from  $A$  to  $A$ .

**Example:**

- Suppose that  $A = \{a, b, c\}$ . Then  $R = \{(a, a), (a, b), (a, c)\}$  is a relation on  $A$ .
- Let  $A = \{1, 2, 3, 4\}$ . The ordered pairs in the relation  $R = \{(a, b) \mid a \text{ divides } b\}$  are  $(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3)$ , and  $(4, 4)$ .

## Binary Relation on a Set (*cont.*)

**Question:** How many relations are there on a set  $A$ ?

**Solution:** Because a relation on  $A$  is the same thing as a subset of  $A \times A$ , we count the subsets of  $A \times A$ . Since  $A \times A$  has  $n^2$  elements when  $A$  has  $n$  elements, and a set with  $m$  elements has  $2^m$  subsets, there are  $2^{|A|^2}$  subsets of  $A \times A$ . Therefore, there are  $2^{|A|^2}$  relations on a set  $A$ .

## Binary Relations on a Set (*cont.*)

**Example:** Consider these relations on the set of integers:

$$\begin{aligned} R_1 &= \{(a,b) \mid a \leq b\}, & R_4 &= \{(a,b) \mid a = b\}, \\ R_2 &= \{(a,b) \mid a > b\}, & R_5 &= \{(a,b) \mid a = b + 1\}, \\ R_3 &= \{(a,b) \mid a = b \text{ or } a = -b\}, & R_6 &= \{(a,b) \mid a + b \leq 3\}. \end{aligned}$$

Note that these relations are on an infinite set and each of these relations is an infinite set.

Which of these relations contain each of the pairs

$(1,1)$ ,  $(1,2)$ ,  $(2,1)$ ,  $(1,-1)$ , and  $(2,2)$ ?

**Solution:** Checking the conditions that define each relation, we see that the pair  $(1,1)$  is in  $R_1, R_3, R_4$ , and  $R_6$ ;  $(1,2)$  is in  $R_1$  and  $R_6$ ;  $(2,1)$  is in  $R_2, R_5$ , and  $R_6$ ;  $(1,-1)$  is in  $R_2, R_3$ , and  $R_6$ ;  $(2,2)$  is in  $R_1, R_3$ , and  $R_4$ .

## Reflexive (自反) Relations

**Definition:**  $R$  is *reflexive* iff  $(a,a) \in R$  for every element  $a \in A$ . Written symbolically,  $R$  is reflexive if and only if

$$\forall x[x \in U \rightarrow (x,x) \in R]$$

**Example:** The following relations on the integers are reflexive:

$$\begin{aligned} R_1 &= \{(a,b) \mid a \leq b\}, \\ R_3 &= \{(a,b) \mid a = b \text{ or } a = -b\}, \\ R_4 &= \{(a,b) \mid a = b\}. \end{aligned}$$

If  $A = \emptyset$  then the empty relation is reflexive vacuously. That is the empty relation on an empty set is reflexive!

The following relations are not reflexive:

$$\begin{aligned} R_2 &= \{(a,b) \mid a > b\} \text{ (note that } 3 \not> 3), \\ R_5 &= \{(a,b) \mid a = b + 1\} \text{ (note that } 3 \neq 3 + 1), \\ R_6 &= \{(a,b) \mid a + b \leq 3\} \text{ (note that } 4 + 4 \not\leq 3). \end{aligned}$$



## Symmetric Relations

**Definition:**  $R$  is *symmetric* iff  $(b,a) \in R$  whenever  $(a,b) \in R$  for all  $a,b \in A$ . Written symbolically,  $R$  is symmetric if and only if

$$\forall x \forall y [(x,y) \in R \rightarrow (y,x) \in R]$$

**Example:** The following relations on the integers are symmetric:

$$R_3 = \{(a,b) \mid a = b \text{ or } a = -b\},$$

$$R_4 = \{(a,b) \mid a = b\},$$

$$R_6 = \{(a,b) \mid a + b \leq 3\}.$$

The following are not symmetric:

$$R_1 = \{(a,b) \mid a \leq b\} \text{ (note that } 3 \leq 4, \text{ but } 4 \not\leq 3),$$

$$R_2 = \{(a,b) \mid a > b\} \text{ (note that } 4 > 3, \text{ but } 3 \not> 4),$$

$$R_5 = \{(a,b) \mid a = b + 1\} \text{ (note that } 4 = 3 + 1, \text{ but } 3 \neq 4 + 1).$$

## Antisymmetric Relations

**Definition:** A relation  $R$  on a set  $A$  such that for all  $a,b \in A$  if  $(a,b) \in R$  and  $(b,a) \in R$ , then  $a = b$  is called *antisymmetric*.

Written symbolically,  $R$  is antisymmetric if and only if

$$\forall x \forall y [(x,y) \in R \wedge (y,x) \in R \rightarrow x = y]$$

- Example:** The following relations on the integers are antisymmetric:

$$R_1 = \{(a,b) \mid a \leq b\},$$

$$R_2 = \{(a,b) \mid a > b\},$$

$$R_4 = \{(a,b) \mid a = b\},$$

$$R_5 = \{(a,b) \mid a = b + 1\}.$$

For any integer, if  $a \leq b$  and  $a \leq b$ , then  $a = b$ .

The following relations are not antisymmetric:

$$R_3 = \{(a,b) \mid a = b \text{ or } a = -b\}$$

(note that both  $(1,-1)$  and  $(-1,1)$  belong to  $R_3$ ),

$$R_6 = \{(a,b) \mid a + b \leq 3\} \text{ (note that both } (1,2) \text{ and } (2,1) \text{ belong to } R_6).$$

## Transitive Relations

**Definition:** A relation  $R$  on a set  $A$  is called transitive if whenever  $(a,b) \in R$  and  $(b,c) \in R$ , then  $(a,c) \in R$ , for all  $a,b,c \in A$ . Written symbolically,  $R$  is transitive if and only if

$$\forall x \forall y \forall z [(x,y) \in R \wedge (y,z) \in R \rightarrow (x,z) \in R]$$

- Example:** The following relations on the integers are transitive:

$$R_1 = \{(a,b) \mid a \leq b\},$$

$$R_2 = \{(a,b) \mid a > b\},$$

$$R_3 = \{(a,b) \mid a = b \text{ or } a = -b\},$$

$$R_4 = \{(a,b) \mid a = b\}.$$

For every integer,  $a \leq b$   
and  $b \leq c$ , then  $a \leq c$ .

The following are not transitive:

$$R_5 = \{(a,b) \mid a = b + 1\} \text{ (note that both } (3,2) \text{ and } (4,3) \text{ belong to } R_5, \text{ but not } (3,3)),$$

$$R_6 = \{(a,b) \mid a + b \leq 3\} \text{ (note that both } (2,1) \text{ and } (1,2) \text{ belong to } R_6, \text{ but not } (2,2)).$$

### Question:

**Symmetric, transitive  $\Rightarrow$  reflexive ?**

$$\left. \begin{array}{l} (a,b) \in R \\ R \text{ is symmetric} \end{array} \right\} \Rightarrow \left. \begin{array}{l} (b,a) \in R \\ R \text{ is transitive} \end{array} \right\} \Rightarrow (a,a) \in R$$

This argument makes an assumption that  $\forall a \exists b (a,b) \in R$

**Therefore, symmetry and transitivity are not enough to infer reflexivity**



## Combining Relations

- Given two relations  $R_1$  and  $R_2$ , we can combine them using basic set operations to form new relations such as  $R_1 \cup R_2$ ,  $R_1 \cap R_2$ ,  $R_1 - R_2$ , and  $R_2 - R_1$ .
- Example:** Let  $A = \{1,2,3\}$  and  $B = \{1,2,3,4\}$ . The relations  $R_1 = \{(1,1), (2,2), (3,3)\}$  and  $R_2 = \{(1,1), (1,2), (1,3), (1,4)\}$  can be combined using basic set operations to form new relations:

$$R_1 \cup R_2 = \{(1,1), (1,2), (1,3), (1,4), (2,2), (3,3)\}$$

$$R_1 \cap R_2 = \{(1,1)\} \quad R_1 - R_2 = \{(2,2), (3,3)\}$$

$$R_2 - R_1 = \{(1,2), (1,3), (1,4)\}$$

## Composition

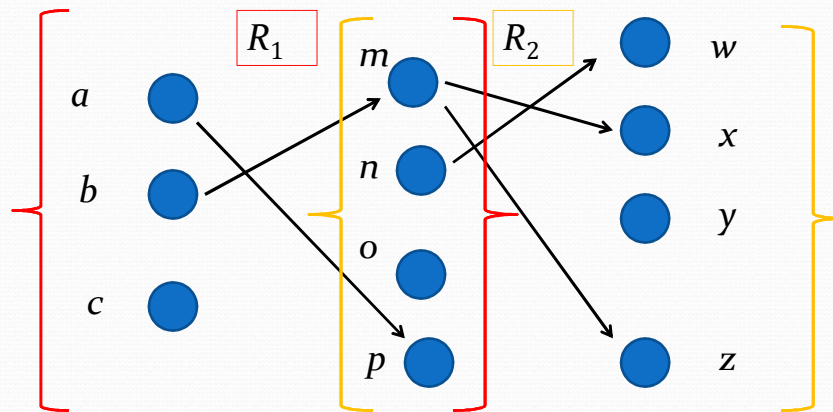
**Definition:** Suppose

- $R_1$  is a relation from a set  $A$  to a set  $B$ .
- $R_2$  is a relation from  $B$  to a set  $C$ .

Then the *composition* (or *composite*) of  $R_2$  with  $R_1$ , is a relation from  $A$  to  $C$  where

- if  $(x,y)$  is a member of  $R_1$  and  $(y,z)$  is a member of  $R_2$ , then  $(x,z)$  is a member of  $R_2 \circ R_1$ .

## Representing the Composition of a Relation



$$R_2 \circ R_1 = \{(b,x), (b,z)\}$$

## relational composition

- Let  $M$  be the relation “is mother of”
- Let  $F$  be the relation “is father of”
- What is  $M \circ F$ ?
  - If  $(a,b) \in F$ , then  $a$  is the father of  $b$
  - If  $(b,c) \in M$ , then  $b$  is the mother of  $c$
  - Thus,  $M \circ F$  denotes the relation “maternal grandfather” (外公)
- What is  $F \circ M$ ?
  - If  $(a,b) \in M$ , then  $a$  is the mother of  $b$
  - If  $(b,c) \in F$ , then  $b$  is the father of  $c$
  - Thus,  $F \circ M$  denotes the relation “paternal grandmother” (奶奶)
- What is  $M \circ M$ ?
  - If  $(a,b) \in M$ , then  $a$  is the mother of  $b$
  - If  $(b,c) \in M$ , then  $b$  is the mother of  $c$
  - Thus,  $M \circ M$  denotes the relation “maternal grandmother” (外婆)
- Note that  $M$  and  $F$  are not transitive relations!!!

## Powers of a Relation

**Definition:** Let  $R$  be a binary relation on  $A$ . Then the powers  $R^n$  of the relation  $R$  can be defined inductively by:

- Basis Step:  $R^1 = R$
- Inductive Step:  $R^{n+1} = R^n \circ R$
- Example:  $R = \{(1,1), (2,1), (3,2), (4,3)\}$

Find  $R^2$ ,  $R^3$ , and  $R^4$

$$R^2 = R \circ R = \{(1,1), (2,1), (3,1), (4,2)\}$$

$$R^3 = R^2 \circ R = \{(1,1), (2,1), (3,1), (4,1)\}$$

$$R^4 = R^3 \circ R = \{(1,1), (2,1), (3,1), (4,1)\}$$

## Example

- $R = \{(a,b), a \text{ is parent of } b \text{ or vice versa}\}$
- $R^2 = \{(a,b), a \text{ is grandparent of } b \text{ or vice versa}\}$
- $N$ -generations blood relationship: if  $(a,b) \in R^n$ , we say  $a$  and  $b$  have  $n$ -generations blood relationship

## Theorem 1

- Then relation  $R$  on a set  $A$  is transitive if and only if  $R^n \subseteq R$ . ( $n=1,2,3,\dots$ )
- **If part:**  $R^n \subseteq R$ ,  $R^2 \subseteq R$ . if  $(a,b) \in R$  and  $(b,c) \in R$  for any  $a,b,c \in A$ , then  $(a,c) \in R^2$ , hence,  $(a,c) \in R$ ,  $R$  is transitive.
- **Only if part:** if  $R$  is transitive,  $(a,c) \in R^2$ , then there exist  $b \in A$  such that  $(a,b) \in R$  and  $(b,c) \in R$ . Hence  $(a,c) \in R$
- This implies that  $R^2 \subseteq R$

## Cont...

- Further more,  $R^3 = R^2 \circ R \subseteq R \circ R = R^2 \subseteq R$
- Then for any  $n=1,2,3,\dots$
- $R^n = R^{n-1} \circ R \subseteq \dots \subseteq R \circ R = R^2 \subseteq R$
- **Inverse Relation:** Let  $R$  be a relation from set  $A$  to set  $B$ , the inverse of  $R$  is a relation from  $B$  to  $A$  such that :
- $R^{-1} = \{(a,b) | (b,a) \in R\}$

## Homework

- 第八版 Sec. 9.1 7(a,c,h), 26, 32, 49, 53

## Representing Relations

Section 9.3

## Section Summary

- Representing Relations using Matrices
- Representing Relations using Digraphs

## Representing Relations Using Matrices

- A relation between finite sets can be represented using a zero-one matrix.
- Suppose  $R$  is a relation from  $A = \{a_1, a_2, \dots, a_m\}$  to  $B = \{b_1, b_2, \dots, b_n\}$ .
  - The elements of the two sets can be listed in any particular arbitrary order. When  $A = B$ , we use the same ordering.
- The relation  $R$  is represented by the matrix  $M_R = [m_{ij}]$ , where

$$m_{ij} = \begin{cases} 1 & \text{if } (a_i, b_j) \in R, \\ 0 & \text{if } (a_i, b_j) \notin R. \end{cases}$$

- The matrix representing  $R$  has a 1 as its  $(i,j)$  entry when  $a_i$  is related to  $b_j$  and a 0 if  $a_i$  is not related to  $b_j$ .



## Examples of Representing Relations Using Matrices

**Example 1:** Suppose that  $A = \{1,2,3\}$  and  $B = \{1,2\}$ . Let  $R$  be the relation from  $A$  to  $B$  containing  $(a,b)$  if  $a \in A$ ,  $b \in B$ , and  $a > b$ . What is the matrix representing  $R$  (assuming the ordering of elements is the same as the increasing numerical order)?

**Solution:** Because  $R = \{(2,1), (3,1), (3,2)\}$ , the matrix is

$$M_R = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}.$$

## Examples of Representing Relations Using Matrices (cont.)

**Example 2:** Let  $A = \{a_1, a_2, a_3\}$  and  $B = \{b_1, b_2, b_3, b_4, b_5\}$ . Which ordered pairs are in the relation  $R$  represented by the matrix

$$M_R = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{bmatrix}?$$

**Solution:** Because  $R$  consists of those ordered pairs  $(a_i, b_j)$  with  $m_{ij} = 1$ , it follows that:

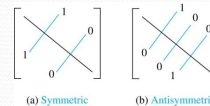
$$R = \{(a_1, b_2), (a_2, b_1), (a_2, b_3), (a_2, b_4), (a_3, b_1), (a_3, b_3), (a_3, b_5)\}.$$

## Matrices of Relations on Sets

- If  $R$  is a reflexive relation, all the elements on the main diagonal of  $M_R$  are equal to 1.



- $R$  is a symmetric relation, if and only if  $m_{ji} = 1$  whenever  $m_{ij} = 1$ .  $R$  is an antisymmetric relation, if and only if  $m_{ij} = 0$  or  $m_{ji} = 0$  when  $i \neq j$ .



(a) Symmetric

(b) Antisymmetric

## Example of a Relation on a Set

**Example 3:** Suppose that the relation  $R$  on a set is represented by the matrix

$$M_R = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

Is  $R$  reflexive, symmetric, and/or antisymmetric?

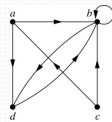
**Solution:** Because all the diagonal elements are equal to 1,  $R$  is reflexive. Because  $M_R$  is symmetric,  $R$  is symmetric and not antisymmetric because both  $m_{1,2}$  and  $m_{2,1}$  are 1.

## Representing Relations Using Digraphs

**Definition:** A *directed graph*, or *digraph*, consists of a set  $V$  of vertices (or *nodes*) together with a set  $E$  of ordered pairs of elements of  $V$  called *edges* (or *arcs*). The vertex  $a$  is called the *initial vertex* of the edge  $(a,b)$ , and the vertex  $b$  is called the *terminal vertex* of this edge.

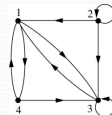
- An edge of the form  $(a,a)$  is called a *loop*.

**Example 7:** A drawing of the directed graph with vertices  $a, b, c$ , and  $d$ , and edges  $(a, b), (a, d), (b, b), (b, d), (c, a), (c, b)$ , and  $(d, b)$  is shown here.



## Examples of Digraphs Representing Relations

**Example 8:** What are the ordered pairs in the relation represented by this directed graph?

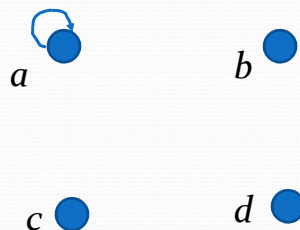


**Solution:** The ordered pairs in the relation are  $(1, 3), (1, 4), (2, 1), (2, 2), (2, 3), (3, 1), (3, 3), (4, 1)$ , and  $(4, 3)$ .

## Determining which Properties a Relation has from its Digraph

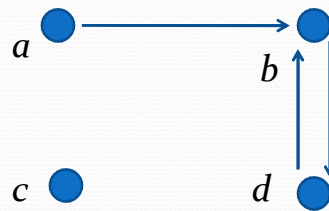
- *Reflexivity*: A loop must be present at all vertices in the graph.
- *Symmetry*: If  $(x,y)$  is an edge, then so is  $(y,x)$ .
- *Antisymmetry*: If  $(x,y)$  with  $x \neq y$  is an edge, then  $(y,x)$  is not an edge.
- *Transitivity*: If  $(x,y)$  and  $(y,z)$  are edges, then so is  $(x,z)$ .

## Determining which Properties a Relation has from its Digraph – Example 1



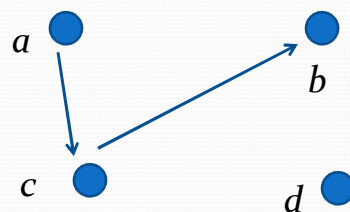
- *Reflexive*? No, not every vertex has a loop
- *Symmetric*? Yes (trivially), there is no edge from one vertex to another
- *Antisymmetric*? Yes (trivially), there is no edge from one vertex to another
- *Transitive*? Yes, (trivially) since there is no edge from one vertex to another

## Determining which Properties a Relation has from its Digraph – Example 2



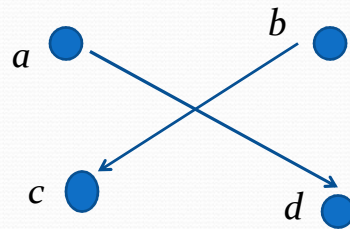
- *Reflexive?* No, there are no loops
- *Symmetric?* No, there is an edge from  $a$  to  $b$ , but not from  $b$  to  $a$
- *Antisymmetric?* No, there is an edge from  $d$  to  $b$  and  $b$  to  $d$
- *Transitive?* No, there are edges from  $a$  to  $c$  and from  $c$  to  $b$ , but there is no edge from  $a$  to  $d$

## Determining which Properties a Relation has from its Digraph – Example 3



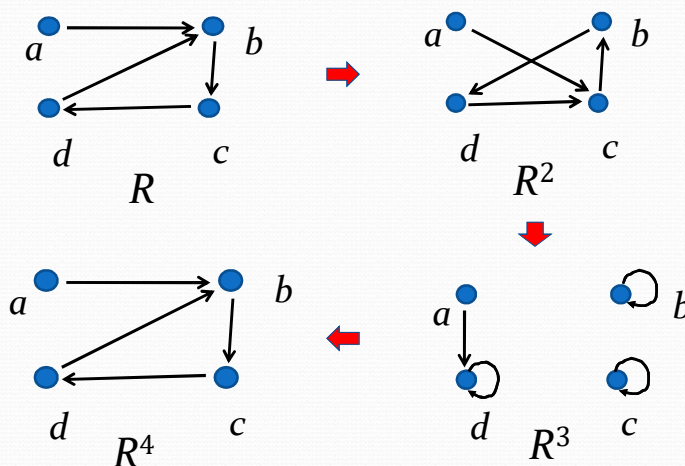
- Reflexive?* No, there are no loops
- Symmetric?* No, for example, there is no edge from  $c$  to  $a$
- Antisymmetric?* Yes, whenever there is an edge from one vertex to another, there is not one going back
- Transitive?* No, there is no edge from  $a$  to  $b$

## Determining which Properties a Relation has from its Digraph – Example 4



- *Reflexive*? No, there are no loops
- *Symmetric*? No, for example, there is no edge from  $d$  to  $a$
- *Antisymmetric*? Yes, whenever there is an edge from one vertex to another, there is not one going back
- *Transitive*? Yes (trivially), there are no two edges where the first edge ends at the vertex where the second edge begins

## Example of the Powers of a Relation



The pair  $(x,y)$  is in  $R^n$  if there is a path of length  $n$  from  $x$  to  $y$  in  $R$  (following the direction of the arrows).



### Inverse relation

$$R = \{(a, b) \mid a \in A, b \in B, aRb\}$$

The **inverse relation** from  $B$  to  $A$ :  $R^{-1}(R^c)$

$$\{(b, a) \mid (a, b) \in R, a \in A, b \in B\}$$

### Question:

How to get  $R^{-1}$  ?

(1) Using the definition directly

For example,  $R = \{(a, b) \mid a \mid b, a, b \in \mathbb{Z}^+\}$   
 $R^{-1} = \{(a, b) \mid b \mid a, a, b \in \mathbb{Z}^+\}$

(2) Reverse all the arcs in the digraph representation of  $R$

(3) Take the transpose  $M_R^T$  of the connection matrix  $M_R$  of  $R$ .

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### The properties of relation operations

Suppose that  $R, S$  are the relations from  $A$  to  $B$ ,  $T$  is the relation from  $B$  to  $C$ ,  $P$  is the relation from  $C$  to  $D$ , then

(1)  $(R \cup S)^{-1} = R^{-1} \cup S^{-1}$

**Proof:**

$$\forall (x, y) \in (R \cup S)^{-1}$$

$$\Leftrightarrow (y, x) \in R \cup S$$

$$\Leftrightarrow (y, x) \in R \text{ or } (y, x) \in S$$

$$\Leftrightarrow (x, y) \in R^{-1} \text{ or } (x, y) \in S^{-1}$$

$$\Leftrightarrow (x, y) \in R^{-1} \cup S^{-1}$$

### The properties of relation operations

Suppose that  $R, S$  are the relations from  $A$  to  $B$ ,  $T$  is the relation from  $B$  to  $C$ ,  $P$  is the relation from  $C$  to  $D$ , then

$$(1) (R \cup S)^{-1} = R^{-1} \cup S^{-1}$$

$$(2) (R \cap S)^{-1} = R^{-1} \cap S^{-1}$$

$$(3) (\bar{R})^{-1} = \overline{R^{-1}}$$

$$(4) (R - S)^{-1} = R^{-1} - S^{-1}$$

$$(5) (A \times B)^{-1} = B \times A$$

*Proof:*

$$\forall (x, y) \in (A \times B)^{-1}$$

$$\Leftrightarrow (y, x) \in A \times B$$

$$\Leftrightarrow (x, y) \in B \times A$$

### The properties of relation operations

Suppose that  $R, S$  are the relations from  $A$  to  $B$ ,  $T$  is the relation from  $B$  to  $C$ ,  $P$  is the relation from  $C$  to  $D$ , then

$$(1) (R \cup S)^{-1} = R^{-1} \cup S^{-1}$$

$$(2) (R \cap S)^{-1} = R^{-1} \cap S^{-1}$$

$$(3) (\bar{R})^{-1} = \overline{R^{-1}}$$

$$(4) (R - S)^{-1} = R^{-1} - S^{-1}$$

$$(5) (A \times B)^{-1} = B \times A$$

$$(6) \bar{R} = A \times B - R$$

$$(7) (S \circ T)^{-1} = T^{-1} \circ S^{-1}$$

$$(8) (R \circ T) \circ P = R \circ (T \circ P)$$

$$(9) (R \cup S) \circ T = R \circ T \cup S \circ T$$

# Homework

Sec. 9.3 13,14,31

## Closures of Relations

Section 9.4

## Definition of Closure

- The *closure* of a relation  $R$  with respect to property  $P$  is the relation obtained by adding the minimum number of ordered pairs to  $R$  to obtain property  $P$ .

## Reflexive Closure

- In terms of the digraph representation of  $R$ :
  - Add loops to all vertices to find the reflexive closure
- In terms of the 0-1 matrix representation:
  - Put 1's on the diagonal to find the reflexive closure
- $r(R) = R \cup \Delta$  where  $\Delta = \{(a, a) | a \in A\}$

## Symmetric Closure

- In terms of the digraph representation of  $R$ :
  - Add arcs in the opposite direction to find the symmetric closure
$$S(R) = R \cup R^{-1}$$
- In terms of the 0-1 matrix representation:
  - Add 1's to the pairs across the diagonals that differ in value



## Transitive Closure

- It is very easy to find the reflexive closure and the symmetric closure, but it is difficult to find the transitive closure
- In terms of the digraph representation of  $R$ :
  - To find the transitive closure, if there is a path from  $a$  to  $b$ , add an arc from  $a$  to  $b$  (can be complicated)

## Transitive Closure

- $R = \{(1,3), (1,4), (2,1), (3,2)\}$  first adding the pairs  $(1,2), (2,3), (2,4), (3,1)$  to  $R$  obtain  $R' = \{(1,3), (1,4), (2,1), (3,2), (1,2), (2,3), (2,4), (3,1)\}$  is not transitive either.
- A path from  $a$  to  $b$  in the digraph  $G$  is a sequence of one or more edges  $(x_0, x_1), (x_1, x_2), \dots, (x_{n-1}, x_n)$  in  $G$  where  $x_0 = a$  and  $x_n = b$ . if  $a = b$ , the path is called circuit or cycle.

## Transitive Closure (Cont.)

- This path is denoted by  $x_0, x_1, x_2, \dots, x_n$  and has **length**  $n$ . the path is called a cycle if it starts and ends at the same vertex.
- Theorem 1: Let  $R$  be a relation on a set  $A$ , there is a path of length  $n$  from  $a$  to  $b$  if and only if  $(a, b) \in R^n$



*Proof:*

① **Inductive basis**

An edge from  $a$  to  $b$  is a path of length 1 which is in  $R^1 = R$ . Hence the assertion is true for  $n = 1$ .

② **Inductive step**

There is a path of length  $n+1$  from  $a$  to  $b$  if and only if there is an  $x$  in  $A$  such that there is a path of length 1 from  $a$  to  $x$  and a path of length  $n$  from  $x$  to  $b$ .

From the Induction Hypothesis,

$$(a, x) \in R \quad (x, b) \in R^n$$

$$(a, b) \in R^n \circ R = R^{n+1}$$

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## Transitive Closure (Cont.)

- $R^* = \bigcup_{i=1}^{\infty} R^i$ , is called the connectivity relation of  $R$ , which consists of the  $(a, b)$  such that there is path from  $a$  to  $b$ .
- Theorem 2: the transitive closure of a relation  $R$  (denoted by  $t(R)$ ) equals the connectivity  $R^*$
- $R^* = \bigcup_{i=1}^{\infty} R^i = t(R)$ .

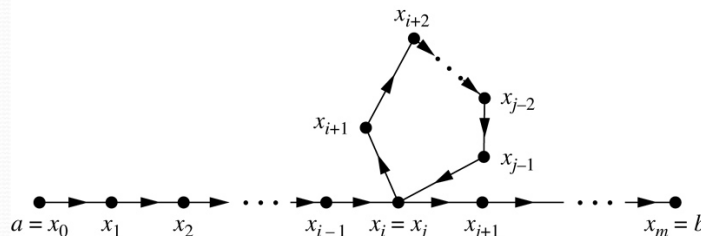
## Proof

- To prove  $R^*$  is transitive closure we must prove:
- (1)  $R^* \supseteq R$ . It is obvious by definition
- (2)  $R^*$  is transitive. If  $(a,b) \in R^*$ ,  $(b,c) \in R^*$ , it implies there is a path from  $a$  to  $b$  and a path from  $b$  to  $c$ , hence there is a path from  $a$  to  $c$  through  $b$ .
- (3)  $R^*$  is minimum. If  $S$  is also a transitive relation containing  $R$ , then  $S \supseteq R^*$ . It is obvious that  $S^* = S$ . since  $S \supseteq R$ , then  $S^* \supseteq R^*$ , hence  $S \supseteq R^*$ .

## Lemma 1

- $A$  is a set containing  $n$  elements.  $R$  is relation on  $A$ . if there is a path from  $a$  to  $b$ , then there is such path with length not exceeding  $n$ . if  $a \neq b$ , there is such path with length not exceeding  $n-1$ .
- From this lemma,  $t(R) = \bigcup_{i=1}^n R^i$

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**FIGURE 2** Producing a Path with Length Not Exceeding  $n$ .

## Transitive Closure (Cont.)

- Theorem 3  $M_{R^*} = M_R \vee M_R^2 \vee M_R^3 \vee \dots \vee M_R^n$

$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} \quad M_R^2 = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} = M_R^3 = M_{R^*}$$

## Cont...

Algorithm 1 A procedure for computing the transitive closure

**procedure** *transitive\_closure* ( $M_R$ : zero-one  $n \times n$  matrix)

$A := M_R$

$B := A$

**for**  $i := 2$  **to**  $n$

**begin**

$A := A \odot M_R$

$B := B \vee A$

**end** {  $B$  is the zero-one matrix for  $R^*$  }

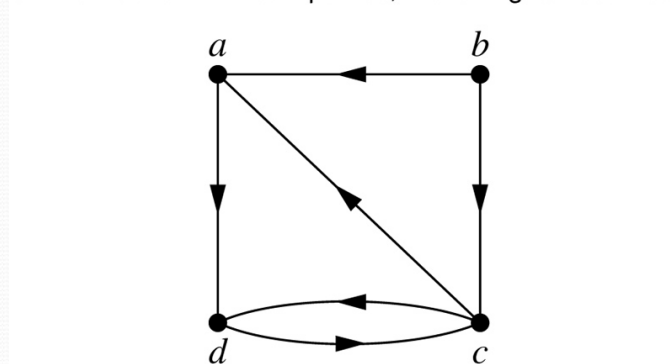
## Transitive Closure (Cont.)

- Warshall's algorithm an efficient method for computing the transitive closure of a relation.
- Interior vertices of a path:  $a, x_1, x_2, \dots, x_{m-1}, b$ .  $x_1, x_2, \dots, x_{m-1}$  are interior vertices
- Matrices:  $M_R = W_0, W_1, W_2, \dots, W_n = M_{R^*}$
- Named after Stephen Warshall in 1960
  - $2n^3$  bit operation
  - Also called Roy-Warshall algorithm, Bernard Roy in 1959
  - Previous algorithm 1 using  $2n^3 (n-1)$  bit operation  

$$n^2(2n-1)(n-1) + (n-1)n^2 = 2n^3(n-1) = O(n^4)$$

## FIGURE 3 (9.4,p604)

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**FIGURE 3** The Directed Graph of the Relations  $R$ .

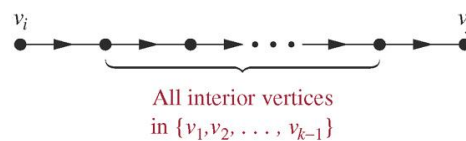
## Warshall's algorithm

- Observation: we can compute  $W_k$  directly from  $W_{k-1}$ 
  - Two cases (Fig. 4)
    - (a) There is a path from  $v_i$  to  $v_j$  with its interior vertices among the first  $k-1$  vertices
      - $w_{ij}^{(k-1)}=1$
    - (b) There are paths from  $v_i$  to  $v_k$  and from  $v_k$  to  $v_j$  that have interior vertices only among the first  $k-1$  vertices
      - $w_{ik}^{(k-1)}=1$  and  $w_{kj}^{(k-1)}=1$

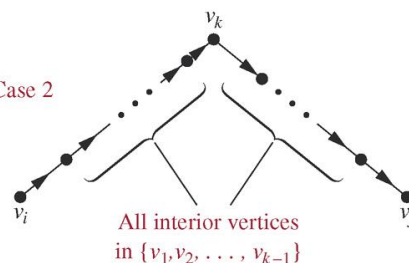
### FIGURE 4 (9.4)

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Case 1



Case 2



**FIGURE 4 p605**

Adding  $v_k$  to the  
Set of Allowable  
Interior Vertices.

## Warshall's algorithm

- Lemma 2: Let  $W_k = w_{ij}^{(k)}$  be the zero-one matrix that has a 1 in its  $(i,j)$ th position iff there is a path from  $v_i$  to  $v_j$  with interior vertices from the set  $\{v_1, v_2, \dots, v_k\}$ . Then  $w_{ij}^{(k)} = w_{ij}^{(k-1)} \vee (w_{ik}^{(k-1)} \wedge w_{kj}^{(k-1)})$ , whenever  $i, j$ , and  $k$  are positive integers not exceeding  $n$ .

## Transitive Closure (Cont.)

- Algorithm 2 warshall algorithm
- Procedure warshall( $M_R: n \times n$  zero-one matrix)
- $W = M_R$
- For  $k=1$  to  $n$
- Begin
- For  $i=1$  to  $n$
- Begin
- For  $j=1$  to  $n$
- $W_{ij} = w_{ij} \vee (w_{ik} \wedge w_{kj})$
- End
- End



【Example】 Let  $A = \{1,2,3,4,5\}, R = \{(1,1), (1,2), (2,4), (3,5), (4,2)\}, t(R) = ?$

*Solution:*

$$\begin{array}{c}
 M = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{k=1} \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{k=2} \begin{bmatrix} 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \\
 \xrightarrow{k=3} \begin{bmatrix} 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{k=4} \begin{bmatrix} 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{k=5} \begin{bmatrix} 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}
 \end{array}$$

$$W_{ij} = w_{ij} \vee (w_{ik} \wedge w_{kj})$$

If  $(w_{ik} == 1)$

$$W_{ij} = w_{ij} \vee w_{kj}$$

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## Homework

Sec.9.4 2, 6, 9(6), 11(6), 20, 28(a), 29